Design & Analysis of Algorithms_2023_PYQ

Group-A (Very Short Answer Type Question)

Answer any ten of the following: [1 x 10 = 10]

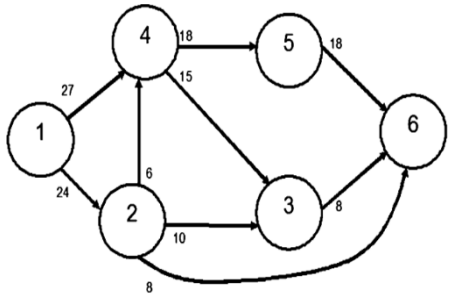
1. State True/False: Approximation algorithm guarantees the best quality of solution.
2. State True/False: O(log n) > O(n)
3. What is the time complexity of Knapsack algorithm using Greedy method?
4. State True/False: Adjacency matrix and Path Matrix are the same.
5. State True/False: Ordered searching algorithm is a Polynomial Algorithm.
6. State True/False: Brute Force method gives the best quality of solution.
7. State True/False: Travelling Salesman Problem is a non-polynomial problem.
8. Name one heuristic method of searching.
9. Best case time complexity of Binary search algorithm is _____
10. The time complexity of Floyd's algorithm is _____
11. Backtracking follows _____ traversal technique.
12. What is spanning tree?

Group-B (Short Answer Type Question)

Answer any three of the following: [5 x 3 = 15]

2. Write down an algorithm to solve the Job Sequencing with Deadline Problem using Greedy method. What is the time complexity of your algorithm?

3. Find the maximum flow of the following network. Mention each step clearly.



4. Write down an algorithm in Greedy method to find the minimum spanning tree by Prim's algorithm. What is the time complexity of your algorithm?

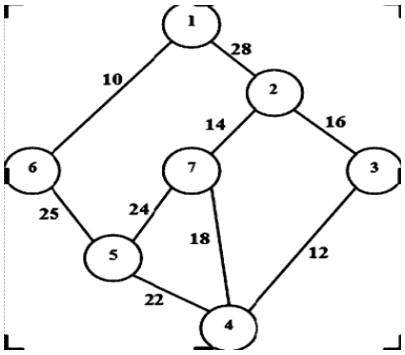
5. (a) State Cook's Theorem. (b) Define Clique with example.

6. Define the following terms: (i) Brute Force Algorithm (ii) Approximation Algorithm

Group-C (Long Answer Type Question)

Answer any three of the following: [15 x 3 = 45]

7. (a) Find minimum spanning tree using Prim's method from the following graph. Mention each step clearly. [10]



(b) Write an algorithm to find all solutions of N-Queens problem using backtracking. [4]

(c) What is the time complexity of your algorithm? [1]

(d) Write an algorithm to find all pairs shortest path using Floyd's method. [4]

8. (a) Write down an algorithm of Quick Sort. [5]

(b) Derive the best, worst and average case time complexity of your algorithm. [2+3+5]

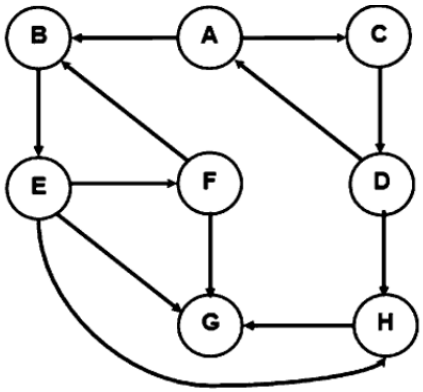
9. (a) Find the minimum number of scalar operations needed to multiply the matrices A1, A2, A3 and A4 having dimensions 30×35, 35×15, 15×5 and 5×10 respectively. [5]

(b) Write down an algorithm using Greedy method to find minimum spanning tree by Kruskal's algorithm. [5]

(c) Find the time complexity of your algorithm. [3]

(d) Differentiate between Backtracking and Branch and Bound. [2]

10. (a) Find the transitive closure of vertex A from the given graph. [3]



(b) Define the following terms:

(i) Directed graph (ii) Undirected graph (iii) In-Degree and Out-Degree (iv) Complete Graph (v) Articulation Point (vi) Bridge [12]

11. (a) Differentiate between recursion and iteration. [3]

(b) Find the time complexity of the following recurrence relations using Master's Theorem:

(i) $T(n) = 2T(n/4) + n^{0.51}$ (ii) $T(n) = \sqrt{2}\, T(n/2) + \log n$ [4+4]

(c) Define little omega notation. Explain with an example. [2+2]